

class08

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```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer (1).csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

#adding the csv file to the quarto document

head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
84348301	0.14250	0.28390	0.2414		0.10520
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419		0.07871	1.0950	0.9053
842517	0.1812		0.05667	0.5435	0.7339
84300903	0.2069		0.05999	0.7456	0.7869
84348301	0.2597		0.09744	0.4956	1.1560
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	
842517	74.08	0.005225	0.01308	0.01860	
84300903	94.03	0.006150	0.04006	0.03832	
84348301	27.23	0.009110	0.07458	0.05661	
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003		0.006193	25.38	17.33

842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			

```
#examining input data

wisc.data <- wisc.df[,-1]
#removing first column

diagnosis <- wisc.df$diagnosis
#setting up vector for diagnosis for later
```

Q1: How many observations are in this dataset?

```
dim(wisc.data)
```

```
[1] 569 30
```

```
#this tells us the number of rows and columns.
nrow(wisc.data)
```

```
[1] 569
```

```
#the number of rows is the number of observations
```

Q2: How many of the observations have a malignant diagnosis?

```
#table function shows us the two values the diagnosis column is seperated into, "M": malignant
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

```
#there are 212 malignant diagnosis
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#using grep to search through column names that have _mean

length(grep("mean", colnames(wisc.data)))
```

```
[1] 10
```

```
#there are 10 variables in the data suffixed with _mean.
```

checking mean and standard deviation

```
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02

smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
#scaling is required because the variables have very different means and standard deviations
```

```
#doing scaling and performing pca
```

```
wisc.pr <- prcomp(wisc.data, scale. = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14

Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4: From your results, what proportion of the original variance is captured by the first principal component (PC1)?

A4: Proportion of original variance in PC1: $0.4427 = 44.27\%$

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

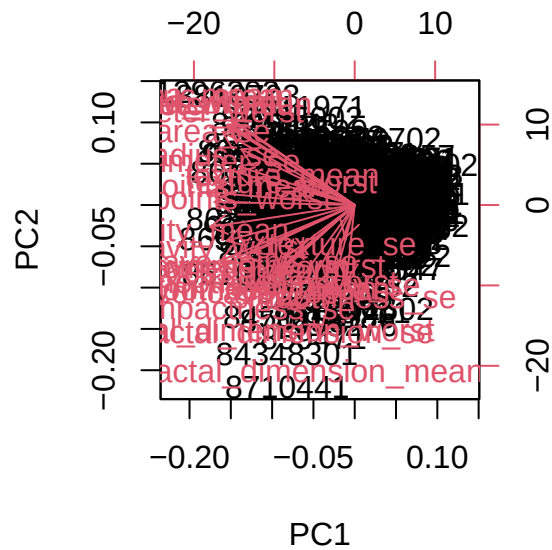
A5: 3 principle componenets are required to describe at least 70% of the original variance in the data (cummulative proportion = 0.72636 for PC3)

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

A6: 7 principle compopents are needed to describe at least 90% of the original variance in the data (PC7= 0.91 cummulative proportion)

```
#creating biplot of wisc.pr using biplot function
```

```
biplot(wisc.pr)
```

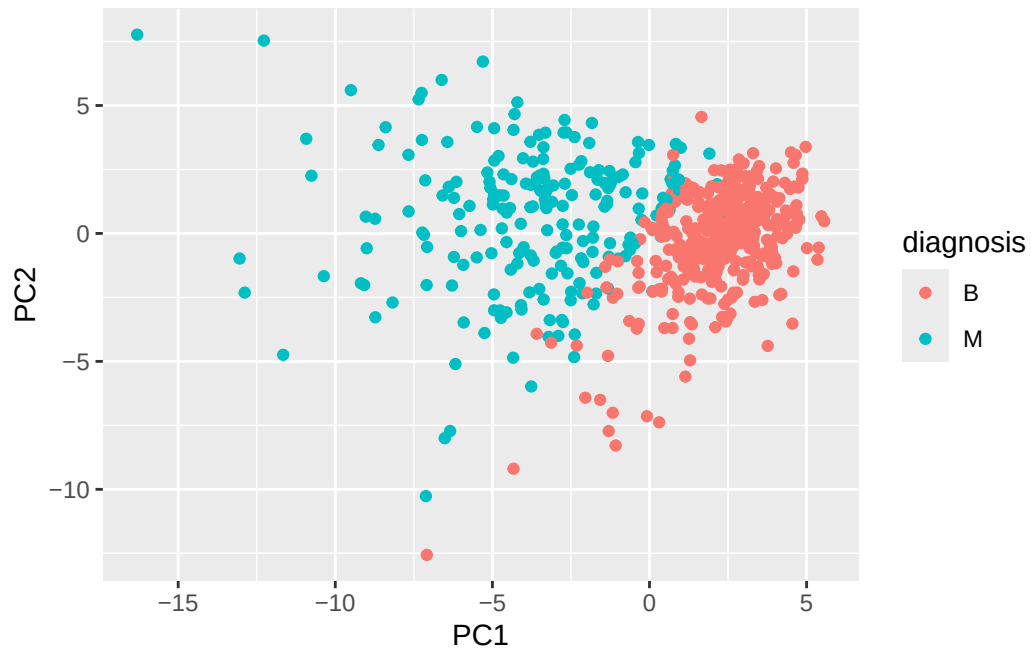


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

A7: This plot is very difficult to understand, it is too small and very crowded. The numbers and words overlap to the point where you can't even really see anything.

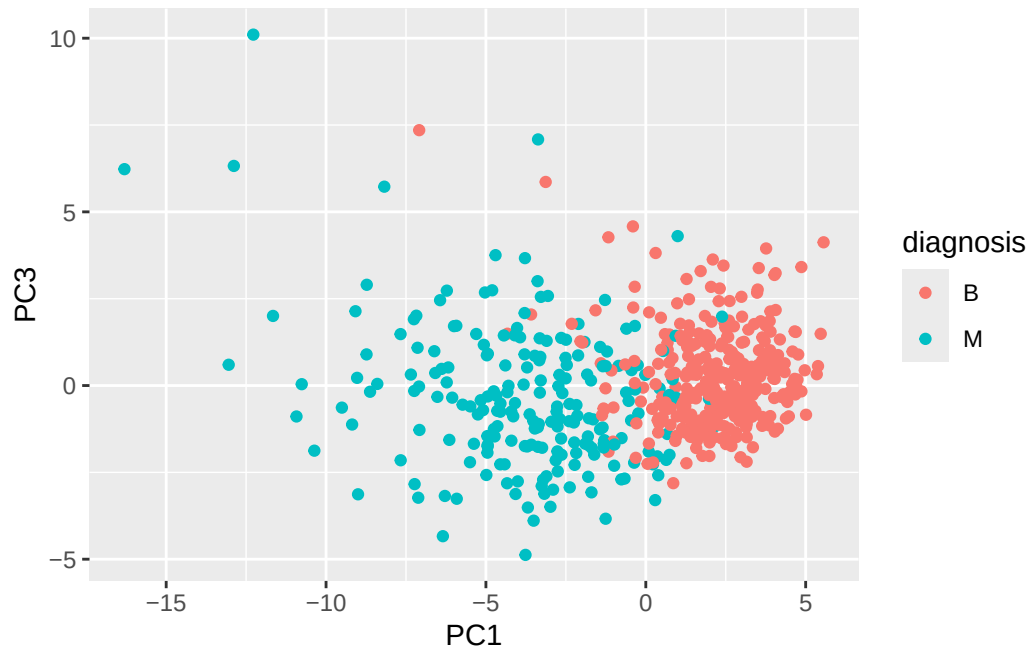
```
#making plot for PC 1 and PC2
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



#I noticed that principle componenet 1 is capturing a seperationg of malignant form benign s

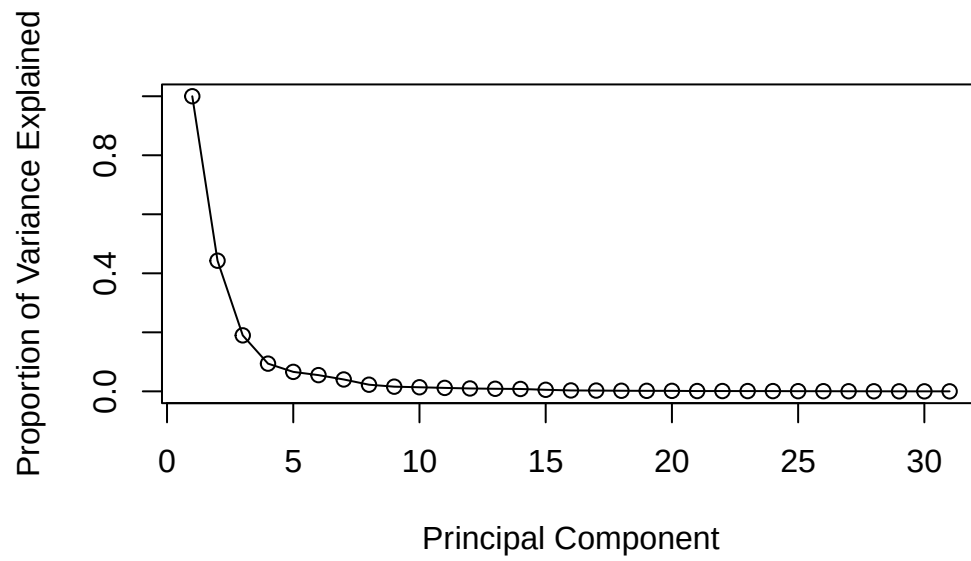
variance stuff

```
#calculating variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

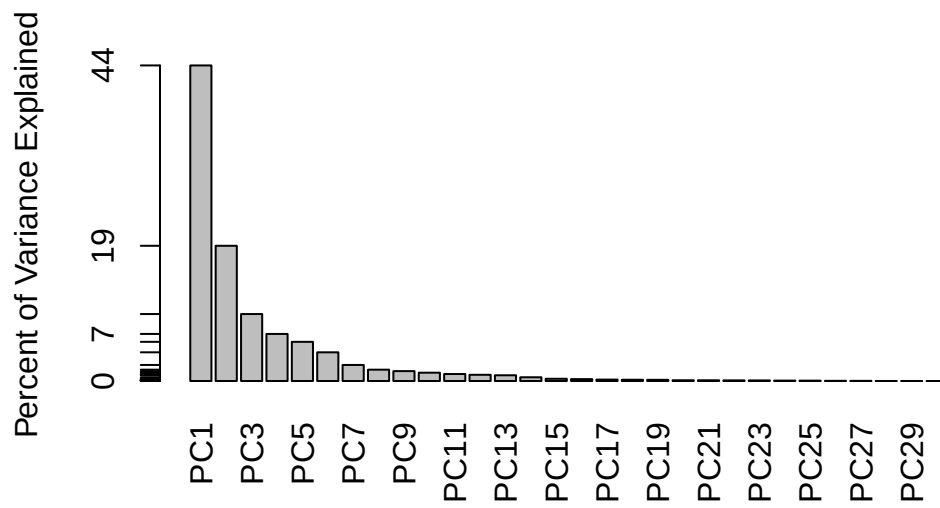
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var/sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Q9: For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

#the loadin =g for concave.points_mean on PC1 is -0.2608538, indicating a moderate contribut

```
wisc.pr$rotation["radius_worst", 1]
```

```
[1] -0.2279966
```

```
wisc.pr$rotation["perimeter_worst", 1]
```

```
[1] -0.2366397
```

```
wisc.pr$rotation["area_worst", 1]
```

```
[1] -0.2248705
```

```
wisc.pr$rotation["radius_mean", 1]
```

```
[1] -0.2189024
```

```
wisc.pr$rotation["perimeter_mean", 1]
```

```
[1] -0.2275373
```

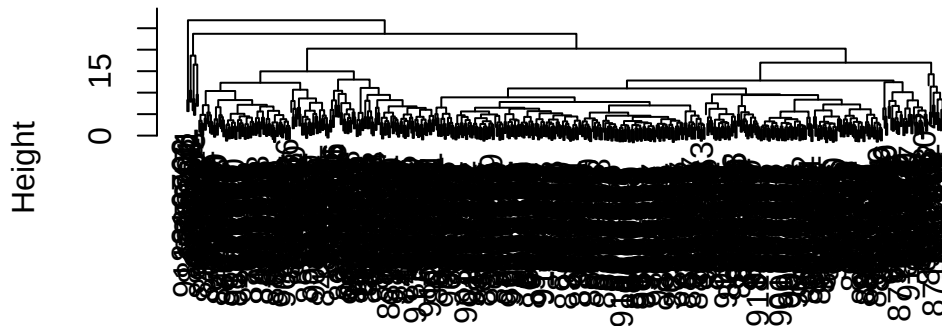
```
wisc.pr$rotation["area_mean", 1]
```

```
[1] -0.220995
```

```
#concave.point_mean seems to have the highest contribution compared to these other features
```

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method="complete")
plot(wisc.hclust)
```

Cluster Dendrogram

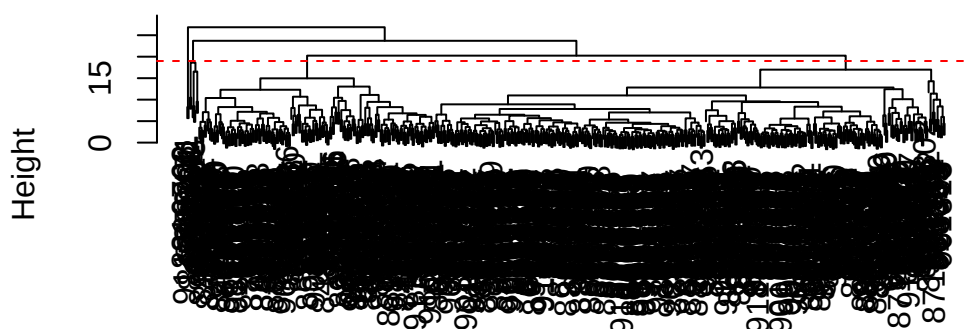


```
data.dist  
hclust (*, "complete")
```

Q10: Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters? plot

```
plot(wisc.hclust)  
abline(h = 19, col="red", lty=2)
```

Cluster Dendrogram



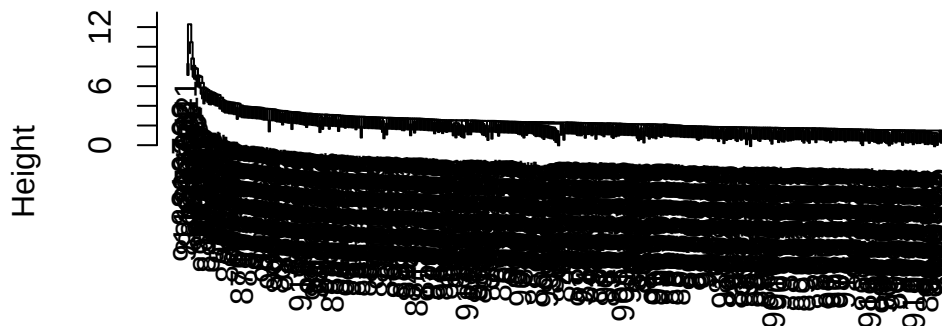
```
data.dist  
hclust (*, "complete")
```

```
#A9: there are 4 clusters are height 19
```

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

```
#trying different methods:  
#trying single  
data.scaled <- scale(wisc.data)  
data.dist <- dist(data.scaled)  
wisc.hclust.single <- hclust(data.dist, method="single")  
plot(wisc.hclust.single)  
abline(h = 19, col="red", lty=2)
```

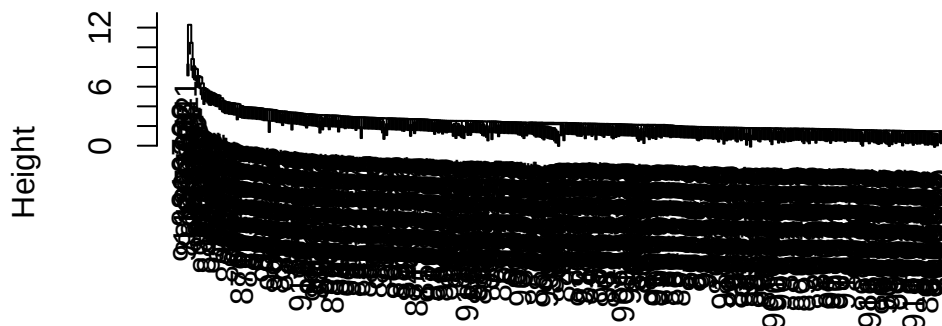
Cluster Dendrogram



```
data.dist  
hclust (*, "single")
```

```
plot(wisc.hclust.single)
```

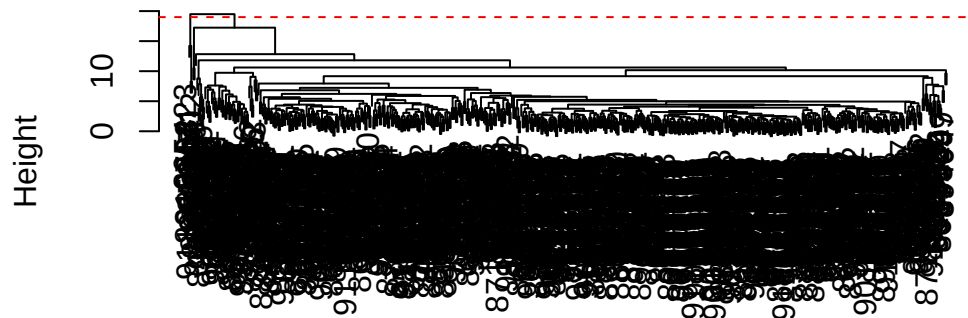
Cluster Dendrogram



```
data.dist  
hclust (*, "single")
```

```
#trying average:
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust.ave <- hclust(data.dist, method="average")
plot(wisc.hclust.ave)
abline(h = 19, col="red", lty=2)
```

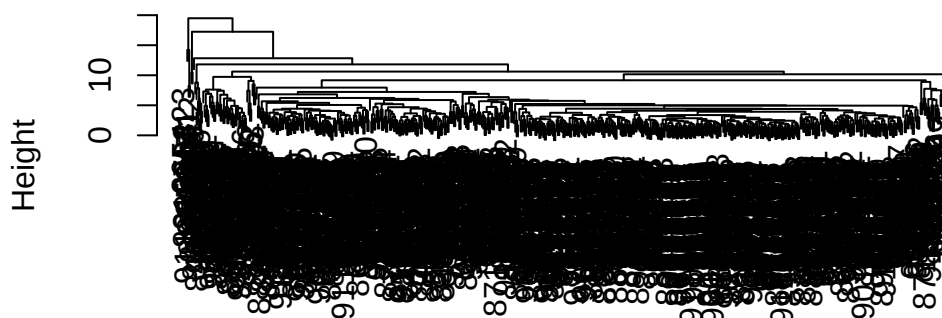
Cluster Dendrogram



```
data.dist
hclust (*, "average")
```

```
plot(wisc.hclust.ave)
```

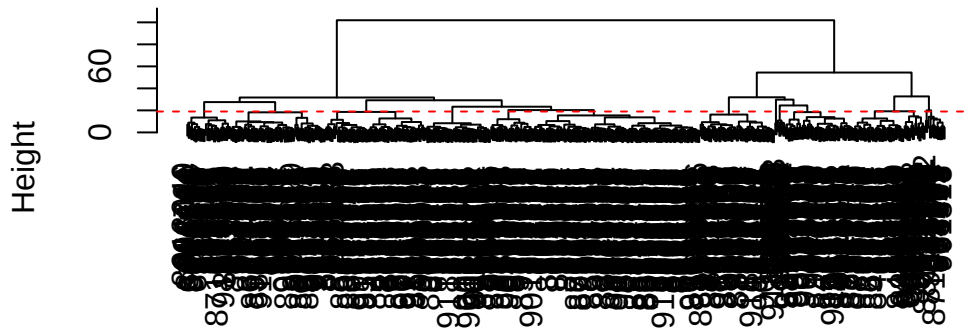
Cluster Dendrogram



```
data.dist  
hclust (*, "average")
```

```
#trying ward.D2  
  
data.scaled <- scale(wisc.data)  
data.dist <- dist(data.scaled)  
wisc.hclust.D2 <- hclust(data.dist, method="ward.D2")  
plot(wisc.hclust.D2)  
abline(h = 19, col="red", lty=2)
```

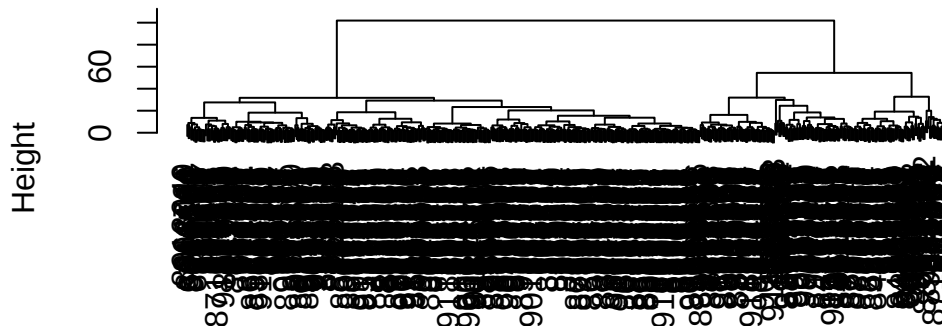

Cluster Dendrogram



```
data.dist  
hclust (*, "ward.D2")
```

```
plot(wisc.hclust.D2)
```

Cluster Dendrogram

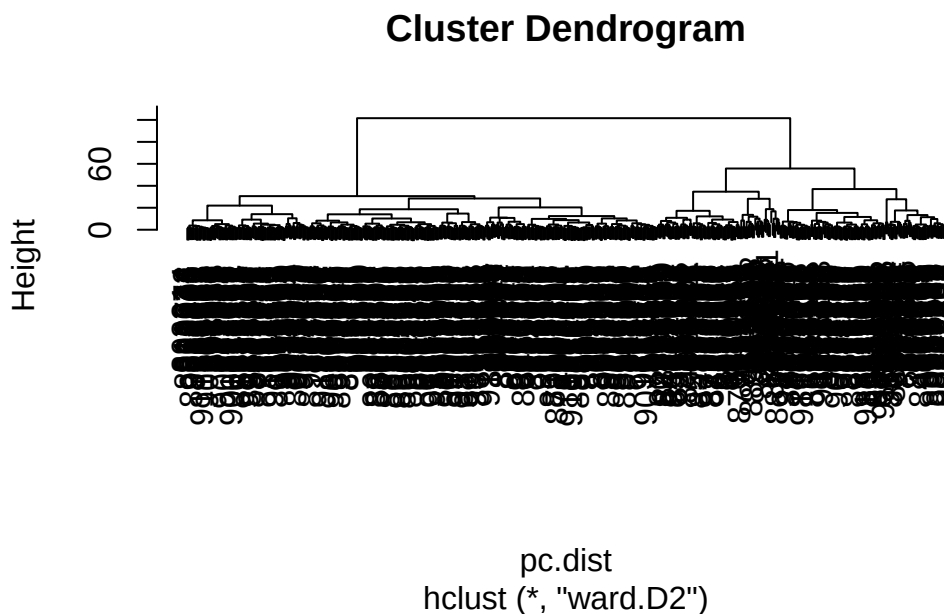


```
data.dist  
hclust (*, "ward.D2")
```

```
#answer12: I think my favorite results are with ward.D2 because it minimizes variance with c
```

clustering on PCA results just following steps from worksheet

```
#creating plot with PC7 (90% variability)
pc90 <- wisc.pr$x[, 1:7]
pc.dist <- dist(pc90)
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```



```
#figuring out what the two clusters are
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
216 353
```

```
table(grps, diagnosis)
```

```

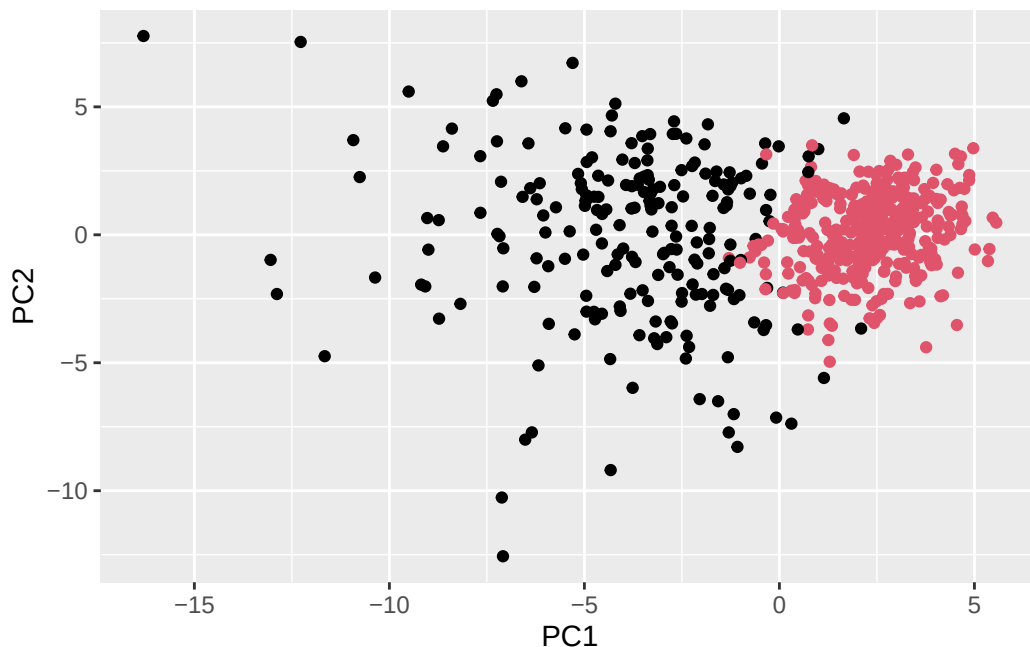
diagnosis
grps   B   M
1    28 188
2    329 24

```

```

ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)

```



```

wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)

```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```

#model without PCA
table(wisc.hclust.clusters, diagnosis)

```

```

diagnosis

```

```
wisc.hclust.clusters  B  M
                     1 12 165
                     2  2  5
                     3 343 40
                     4  0  2
```

```
#model with PCA:
table(wisc.pr.hclust.clusters, diagnosis)
```

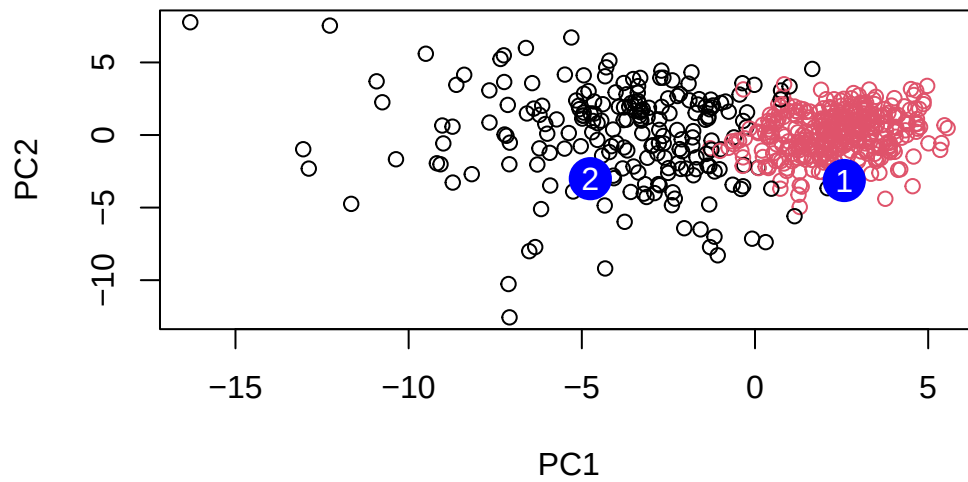
```
              diagnosis
wisc.pr.hclust.clusters  B  M
                     1 28 188
                     2 329 24
```

#A14: This model without PCA performs worse compared to the model with PCA-based clustering.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10      PC11      PC12      PC13      PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
      PC15      PC16      PC17      PC18      PC19      PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
      PC21      PC22      PC23      PC24      PC25      PC26
[1,] 0.1228233 0.09358453 0.08347651 0.1223396 0.02124121 0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
      PC27      PC28      PC29      PC30
[1,] 0.220199544 -0.02946023 -0.015620933 0.005269029
[2,] -0.001134152 0.09638361 0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

A16: I would say patient 1 is at a higher risk and needs to be prioritized because the point for patient 1 is closer to the malignant cluster, which means their cancer is more life threatening. Patient 2 is more near the benign cluster, meaning it is not as urgent compared to patient 1.